

Sidenor

MILL TEST CERTIFICATE



Basauri Plant

SIDENOR ACEROS ESPECIALES S.L.U.
Bo. Ugarte S/n. 48970 Basauri (Vizcaya)

www.sidenor.com

Product Made in Spain

CUSTOMER:MAGELLAN CORPORATION	WORKS REFERENCE:2284560
REFERENCE:216044	SALES ORDER:358337-1
PRODUCT NR:	HEAT NUMBER:643473
MASTER REFERENCE:299496	ROLLED:29.06.2023

REQUIRED PRODUCT			
NITRIDING 135 MOD ROUND BARS AS ROLLED Q+T+STRESS RELIEVED 1.6250" ASTM A-29 (") 17'/19' NORMAL			
EXPEDITION	DELIVERY:0080809580	WEIGHT (KG):7.458	BUNDLES:2
			UNITS:124

MADE ACCORDING TO			
ASTM A29-A29M -20 01.07.2020; ASTM A322 -13(2018)E1 05.2020; ASTM A350-A350M -18 01.05.2018			
ASTM A370 -21 01.11.2021; ASTM E10 -18 01.07.2018; ASTM E381 17 01.06.2017; ASTM E8-E8M -22 01.05.2022			
EN 10204 - OCT. 2004 3.1; GENERAL ELECTRIC B5E1 S22 31.03.2006; MAGELLAN MIC2016 1 30.01.2012			
MIL 86709A-NOT2 - 12.12.1994; SAE AMS 2259 D DIC.2003; SAE-AMS 2301 L 05.2020; SAE-AMS 6470 P 08.2018			
SAE-AMS 6472 G 08.2018; SAE-AMS S6709 A SEPT. 2002			
SIEMENS WESTINGHOUSE PD-SPEC 10305HD SWFC AB 15.04.2005			

CHEMICAL ANALYSIS OF HEAT										U: % HEAT NUMBER:643473
	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	Al
Min.	0,380	0,500	0,200			1,400		0,300		0,950
Max.	0,430	0,700	0,400	0,025	0,025	1,800	0,250	0,400	0,350	1,300
Cor.	0,395	0,516	0,216	0,012	0,007	1,647	0,221	0,346	0,230	1,080
D.I. ASTMA255 (") : 8,266										

JOMINY HARDENABILITY		HARDNESS U.:HRC NORM.TEMP.:982-6/6°C QUENCHING TEMP.:927-6/6°C	
8/16" 12/16"			
Min.	50,0	45,0	
Max.			
Cor.	57,0	56,5	
JOMINY Standard (ASTM A255 10 2014)			

INCLUSIONS (MICROINCLUSIONS)	
Microinclusions std.(ASTM E45 18A 01.06.2018); Type/method(A); A(t):1,5; A(h):0,8; B(t):1,5; B(h):0,8	
C(t):0; C(h):0; D(t):1; D(h):0,5	
INCLUSIONS (MACROINCLUSIONS)	
Standard(SAE-AMS 2301 L 05.2020); Severity(1):Mean value 0,14; Severity(2):Individual value 0,28	
Frequency(1):Mean value 0,15; Frequency(2):Individual value 0,29	

MECHANICAL PROPERTIES AS SUPPLIED (CONDITIONS)	
Temperature of:(1):Quenching 1.688°F; Time:(1):1hours; Cooling:(1):Oil	
Temperature media at the start:(1):. 122°F; Temperature media at the finish:(1):131°F	
Temperature of:(2):Tempering 1.292°F; Time:(2):1hours; Cooling:(2):Water	
Temperature of:(3):Stress Relieved 1.112°F; Time:(3):6hours; Cooling:(3):Air	
MECHANICAL PROPERTIES AS SUPPLIED (TEST)	
Specimen Test location:At 1/2 radius	
Tensile std.(ASTM A370 -21 01.11.2021); Tensile dir.:Longitudinal	
Ts(>=112.000PSI):129.901PSI; Ys(1)(0,2% >=90.000PSI):0,2% 106.248PSI; Ys(2)(0,02% PSI):0,02% 100.969PSI	
El.(4d >=16t):4d 22,2%; Z(>=50%):62,9%	
Hardness Standard(ASTM E10 -18 01.07.2018); Surface hardness:270; Hardness at 1/2 radius(241/285HB):272HB	

ADDITIONAL TESTS (SUPPLY CONDITIONS)	
Decarburization:Total + partial 0,16mm; Grain size std.(ASTM E112 -13(R2021) 01.09.2021)	
Grain size:Austenitic 7; Macroetching std.(ASTM E381 17 01.06.2017); Macroetching(1):# 1	

TECHNOLOGY & QUALITY CERTIFIES THAT THE PRODUCT FULL FILLS THE ORDER'S SPECIFICATIONS

APPROVED BY:Tamar Agustin Martinez

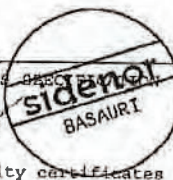
DATE:30.10.2023

REF.:1003375770010

SIGN:

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Analyst of Quality certificates



Customer:BLUE CHIP MACHINE CO | Ord #:T542738 | Part #:508276 | Cust PO #:27084

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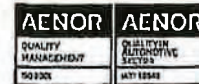
ISO 9001; IATF 16949; ISO 14001; ISO 45001; ISO 50001; ISO 27001 y TI8AX

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SIDENOR ACEROS ESPECIALES S.L.U.
Bo. Ugarte s/n. 48970 Basauri (Vizcaya)

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Product Made in Spain

PO001854
PU 309-212-06-MODOK R

CUSTOMER:MAGELLAN CORPORATION	WORKS REFERENCE:2284560
REFERENCE:216044	SALES ORDER:358337-1
PRODUCT NR:	HEAT NUMBER:643473
MASTER REFERENCE:299496	ROLLED:29.06.2023

Macroetching(2):R 1; Macroetching(3):C 1; Radiactivity:FREE <= 20µroentgens

NON DESTRUCTIVE TESTS

Surface defects std.(ASTM E1444 -22A 01.07.2022)

U.T. standard(ASTM A388-A388M -19 01.09.2019); CRACKS CONTROL 100%;OK; ULTRASONIC INSPECTION 100% : O.K.

ADDITIONAL INFORMATION

Reduction ratio:161,1; Initial Product Format:Ingot; Magnetism:<= 10Gauss

Add. info(1) (Complies SAE-AMS 6472 except pyrometry acc.SAE-AMS 2750)

Add. info(2) (AMS/MIL-S-6709 - Cond. F1)

Add. info(3) (Ultrasonic Inspection Results ("Satisfactory per ASTM A388"))

Add. info(4) ("Steel is free from harmful radioactive contamination")

Material melted and manufactured 100% in Spain through the Electric Arc Furnace (EAF) and Vacuum Degassing route (VOD).

Steel not exposed to Mercury, or to any other metal alloy that is liquid, at ambient temperatures during processing or while in Sidenor's possession.

Material dimensionally controlled, visually inspected, and free from radioactivity (<0,1Bq/g concerning a Co-60).

Steel products were not repaired by welding.

100% anti mix test performed.

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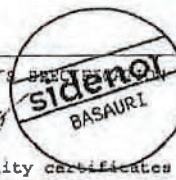
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Analyst of Quality certificates



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1650 Lake Cook Road Suite 300A Deerfield, IL 60015 USA

Telephone: (847) 205-1155

MOR: 700765

SR: 20289855

Packing List / Certificate of Conformance

Sold To: EMJ-Schaumburg

Producing Mill: Sidenor Aceros Especiales S.L.

Customer PO / Item	Bundle #	Description	Heat # / Lot #	Pieces	Weight Pounds
P326887-423 / 508276	986005	NITR-HRQTS-BRRD-1.6250	643473 / 2284560	64	8,206
				Total	64 - 8,206

PO 0018554
PN 309-212-16-MOOLK-RO